Is a Little Slip Worth It? Certified Fast Lifting for Residual Policies Under a Static-to-Kinetic Friction Drop

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Code, data and protocol: https://github.com/CAN-Aidoo/lift-shield

Abstract

A gripper that lifts an object of unknown weight and friction must choose how hard to squeeze and how fast to move. Squeezing less and moving faster are both attractive, and a few millimetres of slip in the fingers is often harmless, which suggests letting a learned policy trade a little slip for a gentler, faster lift. We test that idea in MuJoCo simulation with explicit stick–slip friction. We derive a closed-form bound on slip during a minimum-jerk lift from a probed breakaway grip and an assumed bound $r_{\max}$ on the static-to-kinetic friction ratio, and use it as a shield that raises the grip or slows the lift until the bound is at most 3.5 mm. A linear residual policy trained with evolution strategies cut lift time from 1.00 s to 0.37 s against a quasi-static model controller, at the cost of 35% more grip. Without the shield it learned to grip about 6% below the model, which exploits the model's conservatism at short lift times and fails once friction drops on breakaway: 12–16% of its lifts slipped more than 5 mm for $r \geq 1.1$, against none for the model. With the shield, at training or only at test time, none of 3,520 certified lifts with $r_{\max} \geq 1.2$ exceeded 5 mm, including lifts at $r = 1.3 > r_{\max} = 1.2$, and all results replicated on a fresh confirmation set. The answer to the title is mostly no: once friction may drop by 10% or more, any certified lift is one in which slip never starts, and allowing up to 10 mm of slip saves no grip. An allowance pays (6% less grip) only if friction is known not to drop, and assuming so caused 60 failures in 800 lifts when it did.

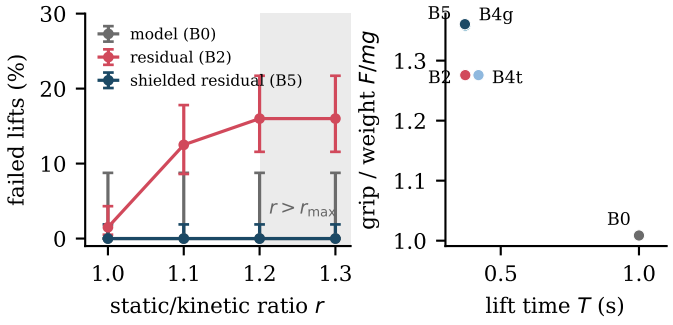


Figure 1. Test set, 40 objects. Left: lifts that slipped more than 5 mm (95% Wilson intervals; residual methods pool 5 training seeds). The unshielded residual (B2) fails once kinetic friction is lower than static ($r > 1$); the shield-trained residual (B5) and the model controller (B0) never do, even beyond the shield's assumed $r_{\max} = 1.2$. Right: mean lift time and grip. All residual variants lift 2.7× faster than the model and grip harder; the shield's price is a further 6% grip (B4g, B5) or 13% time (B4t).

I. Introduction

People lift objects with a grip only modestly above the force at which they would slip, scaled to the friction they feel, and correct it reflexively when the object starts to slip [1]. Robot grippers usually do the opposite: they squeeze hard, because the object's weight and friction are unknown and a dropped object is costly. A light grip matters for fragile, deformable or food items, and a fast lift matters for throughput. Slip detection and grip stabilisation [2, 3] make light grips safer by reacting to slip, but reaction comes after the fact.

In earlier work we studied a failure mode that makes light grips risky: when a grasped object breaks away, friction drops from its static to its kinetic level, and a quasi-static model that ignores the drop under-predicts how far the object travels [4, 5]. Physics shields, which project a learned action onto a set certified by a friction model, prevented this in pivoting and controlled sliding [5]. Lifting is the most common grasp task and the natural next test, and it raises a question the earlier tasks did not: if a few millimetres of slip are acceptable, can a policy deliberately allow them and gain a lighter or faster lift, with a certificate that the slip stays small?

We answer this in simulation with a pre-registered study. Our contributions are:

- A closed-form slip bound for a lift at constant grip (Proposition 1) that needs only a probed breakaway grip and an assumed bound $r_{\max}$ on μ_s/μ_k, used as a shield that projects either the grip or the lift duration.
- Evidence (**Q1**, **Q2**) that an unshielded residual policy learns to exploit the quasi-static model's conservatism, which is safe without a friction drop and fails 12–16% of lifts with one; the shield removes every failure at a cost of 6% grip and no speed.
- A comparison of grip and time projection (**Q3**), certificate checks under $r_{\max}$ mismatch (**Q4**), and an exploratory analysis showing that the bound is a cliff: for $r_{\max} \geq 1.1$ a slip allowance buys essentially no grip. Lift-with-slip only pays when friction is known not to drop.

II. Related Work

Grip force during lifting. Johansson and Westling showed that humans set grip force to the object's friction with a small safety margin, and correct it reflexively when slip begins [1]. Robotic

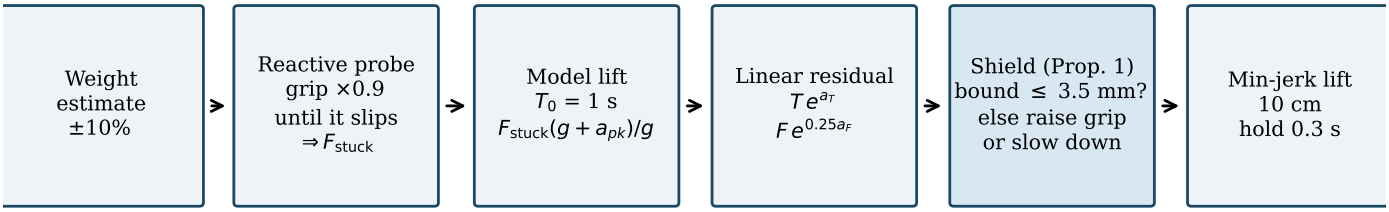


Figure 2. Episode pipeline. The probe and the model lift come from contact mechanics; the residual adjusts the lift time and grip; the shield checks Proposition 1 and, if the slip bound exceeds 3.5 mm, raises the grip (grip axis) or lengthens the lift (time axis) until it does not, declining if neither suffices.

analogues detect or predict slip with tactile sensing and raise the grip in response [2]; Romeo and Zollo survey slip detection methods and sensors [3]. These approaches react to slip. We instead choose the grip and the motion before the lift so that slip is bounded in advance, using only a probe of the breakaway grip.

Residual learning and shields. Residual policies learn corrections to a base controller [6, 7]. Shielding overrides unsafe actions using a model [8]; safety filters based on barrier functions and reachability generalise it [9, 10]. Our previous paper built shields from the static-to-kinetic friction transition for pivoting and sliding [5]. Here the transition happens during a commanded acceleration profile, the shield has two projection axes, and the task explicitly tolerates some slip.

Motion profiles and friction modelling. We lift along a minimum-jerk trajectory [11], which fixes the peak acceleration for a given duration. Stick–slip is emulated in MuJoCo [12] by switching the friction coefficient, as in [5]; richer friction models include rate dependence and pre-sliding [13].

III. Certified Lifting

Task. A parallel gripper holds a rod of mass $m \sim U[0.05, 0.2]$ kg and static friction $\mu_s \sim U[0.4, 0.8]$; kinetic friction is $\mu_k = \mu_s / r$ with r unknown. The gripper lifts the rod 10 cm along a minimum-jerk profile of duration T, whose peak acceleration is $a_{pk} = 5.77\,H/T^2$ [11], at constant grip F. A lift succeeds if the rod slips no more than $S_{\text{tol}} = 5$ mm relative to the fingers.

Probe. With the palm still, the grip starts at $\hat{m}g/(2 \cdot 0.3)$ and is multiplied by 0.9 every 0.4 s until a slip monitor (noise 0.5 mm) reads more than 2.5 mm. The grip then snaps to 2.5× the grip that moved, within 20 ms. F_{stuck}, the last grip that held, is an upper bound on the breakaway grip at rest, so gF/F_{stuck} is a lower bound on the static friction capacity per unit mass at grip F. The weight estimate $\hat{m}$ is within ±10%.

Model controller. The quasi-static Coulomb lift uses $T_0 = 1$ s and $F = F_{\text{stuck}}(g + a_{pk})/g$: just enough static friction at peak acceleration.

Proposition 1. *Let the palm follow an acceleration profile $a(t)$ at constant grip F, and let $r \le r_{\max}$. Slip cannot begin while $g + a(t) \le gF/F_{\text{stuck}}$. From the first instant t_0 where $g + a(t_0) > gF/F_{\text{stuck}}$, define $\bar{a}(t) = g + a(t) - gF/(r_{\max} F_{\text{stuck}})$, $\dot{\bar{v}} = \bar{a}$ with $\bar{v}$ clipped at 0, $\bar{v}(t_0) = 0$. Then slip during the profile is at most $\bar{s} = \int_{t_0} \bar{v}\, dt$.*

Table 1. Rig validation (development objects only).

Check	Result
Static breakaway / $mg/(2\mu)$ ($n = 20$)	0.989 (0.986–1.014)
Lift breakaway / $m(g + a_{pk})/(2\mu)$, $T = 0.25$ / 0.5 / 1 s	0.890 / 0.952 / 0.983
Lifts with zero bound that slipped ($n = 67$)	0
Max excess of slip over a non-zero bound	0.20 mm
Lifts > 5 mm with bound ≤ 5 mm ($n = 120$)	0
Elastic creep at grip margin 1.1–2.0	0.09–0.19 mm

Sketch. The static capacity $2\mu_s F/m = gF/F_b \ge gF/F_{\text{stuck}}$ since the true breakaway grip $F_b \le F_{\text{stuck}}$. While sliding, the kinetic capacity is $gF/(rF_b) \ge gF/(r_{\max} F_{\text{stuck}})$, so the rod's acceleration relative to the fingers is at most $\bar{a}$; re-sticking only lowers it. Comparison of the two velocity equations gives $v \le \bar{v}$. □

Shield. A command (T, F) is certified if $\bar{s} \le s_{\text{lim}} = S_{\text{tol}} - 1.5$ mm, where 1.5 mm is a sensing margin; $\bar{s}$ is evaluated over the 0.6 s grip ramp and settle, the lift and a 0.3 s hold. The *grip projection* raises F to the smallest certified grip; the *time projection* lengthens T to the shortest certified duration (up to 2 s). If neither exists the shield declines and the model's lift is used. We use $r_{\max} = 1.2$.

Residual policy. A linear policy maps six probe features (log breakaway grip over weight, probe resolution, weight, probe count, grip scale, bias) through tanh to $(a_T, a_F) \in [-1, 1]^2$, setting $T = T_0 e^{a_T}$ (clipped to [0.25, 2] s) and $F = F_{\text{model}}(T)\, e^{0.25 a_F}$. The reward is -10 for a failed lift and otherwise $-T/T_0 - 0.1\,F/(mg)$.

IV. Experimental Setup

Simulation. MuJoCo 3 at 1 ms, palm driven kinematically along a vertical slide, spherical pads with elliptic cones and no-slip iterations, rod on a vertical slide joint (a free rod rocked about the grasp axis after slip; we study axial slip only). Friction switches to μ_k when the relative speed exceeds 2 cm/s and back below 2 mm/s. Before any training we validated the rig (Table 1). The static breakaway grip matched $mg/(2\mu)$ to within 1.1%. During lifts the quasi-static model over-predicted the grip needed by 2–11%, more at short T. In 120 random lifts no lift with a bound of zero slipped beyond elastic creep (at most 0.2 mm over the bound), and none with a bound under 5 mm slipped more than 5 mm.

Methods. B0: model controller. B0S: model with the grip

Table 2. Test set, $r_{\max} = 1.2$. Lift time and grip are means over all lifts (they do not depend on r); failures are lifts with slip > 5 mm out of 200 (40 for B0, B0S). None of the failures dropped the rod.

	T (s)	F/mg	failures at $r =$				max slip
			1.0	1.1	1.2	1.3	(mm)
B0 model	1.000	1.009	0	0	0	0	0.49
B0S model+shield	1.000	1.009	0	0	0	0	0.49
B2 residual	0.372	1.276	3	25	32	32	56.0
B4g test shield, grip	0.372	1.358	0	0	0	0	0.12
B4t test shield, time	0.420	1.276	0	0	0	0	0.14
B5 trained shield	0.370	1.361	0	0	0	0	0.11

shield. B2: residual trained and tested without a shield. B4g, B4t: B2's weights with the grip or time shield at test only. B5: residual trained and tested with the grip shield. Residuals were trained with antithetic evolution strategies as in [5] (6 pairs, 6 objects per iteration, $\sigma = 0.1$, Adam step 0.06, 20 iterations, 1,560 episodes per run), 5 seeds each, on $r \sim U[1.0, 1.2]$, starting from zero weights, i.e. from the model controller.

Evaluation and statistics. Test set: 40 new objects, each at $r \in \{1.0, 1.1, 1.2, 1.3\}$. Confirmation set: 40 further objects with $r \sim U[1.0, 1.3]$, run once after the test results were committed. Learned methods are averaged over seeds per object; differences from B0 use paired Wilcoxon tests (lift time, grip) and exact McNemar tests (objects with any failure), Holm-corrected, with hierarchical bootstrap 95% intervals (seeds, then objects). The protocol was committed before training; three dated deviations are recorded in it (Section VI).

V. Results

***Q1**: How much faster, and at what grip?*

Every residual variant lifted 2.7 times faster than the model: B5 cut the lift time by 0.630 s (95% CI 0.629–0.630; Holm $p < 10^{-10}$ at every r), from 1.00 to 0.37 s (Table 2). All ten policies ended at the same corner, $a_T = -1$, the edge of the action range, so the speed gain is set by our action bound, not by the physics. Faster lifts need more grip, because peak acceleration grows as $1/T^2$: B5 gripped $0.352\,mg$ harder than the model (CI 0.330–0.374), 35% more. The unshielded residual gripped less, $+0.267\,mg$, because it set the grip on average 6% (range -11% to $+3\%$) below the model's grip for its own lift time. Lift time is not the whole episode: the probe took a median of 4.1 s (7.5 probe steps), so the end-to-end episode is about 10% shorter.

***Q2**: Does the residual stay safe as friction drops?*

No. B2's grip cut works when $r = 1$ (3 failures in 200 lifts), because the quasi-static model over-predicts the grip needed at short lift times by about 8% (Table 1) and the probe's 0.9 step adds margin. When the rod does start to slip, kinetic friction is too low to stop it: failures rose to 25, 32 and 32 of 200 at $r = 1.1$, 1.2 and 1.3, with slips up to 56 mm. Per object, B2 failed on 14 objects where the model did not at $r = 1.1$ and 16 at $r = 1.2$ and 1.3, never the reverse (Holm $p = 0.001$ and $p = 0.0004$). How much each seed cut differed widely: at $r = 1.2$ the five

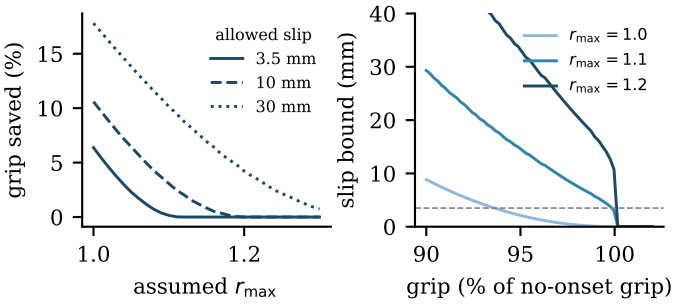


Figure 3. Exploratory: why a slip allowance buys little. Left: grip saved by allowing 3.5, 10 or 30 mm of slip, at $T = 0.37$ s, relative to the grip at which slip never starts, as a function of the assumed $r_{\max}$. Right: the slip bound against grip. With $r_{\max} = 1$ the bound grows gradually as grip falls; with a friction drop it jumps to centimetres as soon as slip can begin.

B2 policies failed 2, 6, 14, 2 and 8 times in 40. All shielded methods had zero failures at every r. On the confirmation set B2 failed 35 of 200 lifts (1, 16 and 18 in the r bands $(1, 1.1]$, $(1.1, 1.2]$ and $(1.2, 1.3]$; 15 objects worse than the model, none better, $p = 6 \times 10^{-5}$), while B4g and B5 failed none, with the same speed (0.371 s, 0.370 s) and grip ($+0.32\,mg$) as on the test set.

***Q3**: Raise the grip or slow the lift?*

Applied to the same B2 commands, both projections intervened on 95.5% of lifts and removed every failure. The grip projection kept B2's speed and added 6.4% grip (1.358 vs $1.276\,mg$); the time projection kept B2's grip and slowed the lift by 13% (0.420 vs 0.372 s). Under our reward, which values 1 s of lift time like $10\,mg$ of grip, the grip projection is better, and training with it in the loop (B5) gave the same result as adding it at test time (B4g): 0.370 vs 0.372 s and 1.361 vs $1.358\,mg$. The shield never declined.

***Q4**: Does the certificate hold, and what if $r_{\max}$ is wrong?*

Every certified lift at $r_{\max} = 1.2$ had a slip bound of exactly zero: the shield placed the grip at the point where slip cannot start. Measured slip was 0.05–0.49 mm, which is elastic creep in the pads, not friction slip; it exceeds a zero bound, so the pre-registered "exceedance" rate is 100%, but the largest excess is a third of the 1.5 mm margin and no certified lift came within 4.5 mm of the tolerance (0 of 2,560 at $r_{\max} = 1.2$, including 640 lifts at $r = 1.3 > r_{\max}$, and 0 of 960 at $r_{\max} = 1.4$). Lifts beyond $r_{\max}$ were safe because slip never started, so the kinetic level never mattered. Underestimating the drop was not safe: with $r_{\max} = 1.0$ the shield let B5 grip 6.4% less ($1.274\,mg$), and 60 of its 600 lifts at $r > 1$ slipped more than 5 mm (10, 25 and 25 at $r = 1.1$, 1.2, 1.3; worst excess over the bound 18 mm), against 0 of 200 at $r = 1$.

The exploratory analysis in Fig. 3 explains this. Allowing 3.5 mm of slip saves 6.4% grip if friction does not drop, 2.3% if it may drop by 5%, 0.15% at 10% and nothing from 15%; allowing 10 mm saves nothing from $r_{\max} = 1.2$. Once slip starts, the kinetic deficit $(g + a)(1 - 1/r_{\max})$ acts for the rest of the acceleration phase, so the slip is either zero or large. The ratio

of certified grip to the no-onset grip was the same for all 40 test objects, since it depends only on $r_{\max}$ and T.

VI. Limitations

All results are from simulation. Stick–slip is emulated by switching the friction coefficient, with thresholds we tuned; real contacts show rate-dependent friction, pre-sliding and pad wear [13], and the rod was constrained to slide axially. The certificate assumes F_{stuck} bounds the breakaway grip during the lift, which held here because the quasi-static model is conservative at short T (Table 1); on hardware this needs checking. The residual's speed gain is bounded by its action range, and the lift is a small part of an episode dominated by a 4 s probe; a faster probe is the larger opportunity. Three deviations from the protocol are recorded in it: the exceedance metric counts elastic creep (we also report the size of the excess); the confirmation set, whose r varies per object, was analysed pooled; and the slip-allowance analysis was exploratory. Training used 1,560 episodes and linear policies; richer policies might find other trade-offs.

VII. Conclusion

A learned lift policy, given the freedom to grip lighter than a friction model, learned to exploit the model's conservatism, which is harmless only when friction does not drop on breakaway. A closed-form slip bound used as a shield removed every failure in simulation, at 6% extra grip and no loss of speed, even beyond its assumed friction drop. Allowing a little slip turned out to be worth little: with a friction drop of 10% or more, the only certified lifts are those in which slip never starts. The next steps are hardware validation of the probe and the bound, and a faster probe.

Data Availability

Code, protocol with deviations, per-episode results and the scripts behind every number: `https://github.com/CAN-Aidoo/lift-shield`.

Use of AI Tools

Code, analysis and drafting were carried out with the assistance of an AI model (Claude, Anthropic) under the author's direction. The author takes responsibility for the content.

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